Porting the miRNA-Pressure Machine to Colorectal Cancer

Literature landscape and where the machine wins

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0. What the machine is (the lens that makes the literature gap legible)

The `mirna_hallmark` engine is a **potential × realized** framework, not a biomarker-panel finder. It separates **pressure** (evidence-weighted, expression-scaled *potential* repression over miRNA→target edges, modeled at five nested resolutions and rolled up over the 50 MSigDB Hallmark programs) from **coupling** (confounder-adjusted partial-Spearman *realization*), tests realization at the **protein layer**, classifies movement along a **healthy→NAT→tumour trajectory** into brake-release / acquired-realized classes, and has a **compartment/EV extension** (the DCIS arc, MH-48..56). That arc's own closing verdict was that it was blocked by exactly two things: *no* *within-patient stage pairs* and *no compartment-resolved small-RNA*. Hold that thought — it is the whole thesis.

The entire CRC literature below is, by contrast, **descriptive differential expression plus correlational biomarker panels.** That mismatch is the opportunity.

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1. The CRC literature landscape, by axis

1.1 The premalignant ladder is real, dual, and exhaustively *described* — but never modeled as realization

CRC is the textbook multistep cancer, and unlike breast it has a genuinely **sampled premalignant ladder** with two molecularly distinct routes:

- **Conventional adenoma–carcinoma sequence** (APC/WNT → KRAS → p53), and
- **Serrated pathway** (BRAF-mutant, CIMP-high, frequently MSI, right-sided) (serrated pathway hallmarks; proximal/BRAF/MSI link).

The miRNA dynamics across these steps are profiled in depth and, importantly, **diverge by pathway**: in the adenoma lineage miR-20a/miR-21/miR-181b rise progressively and miR-93 increases stepwise, whereas the serrated lineage shows selective downregulation/plateauing (Murakami 2025, serrated vs AD miRNA dynamics). The "multi-hit" framing — different miRNAs switching at normal→adenoma vs adenoma→carcinoma — is established (Oberg, PLoS One; normal-adenoma-carcinoma transition; polyp landscape). Specific drivers are mechanistically nailed: the **miR-17-92 cluster** at 13q is MYC-driven, rises across adenoma→carcinoma with 13q gain, and forces a carcinoma-like signature when overexpressed in adenoma organoids, with **miR-92a** the dominant component (13q/MYC; organoid); **miR-31** tracks the BRAF/serrated/CIMP axis (EZH2/miR-31 serrated); **miR-34a** is a p53 target and is methylation-silenced, substituting for p53 loss (miR-34a methylation/metastasis); **miR-143/145** is repressed by oncogenic KRAS in a feed-forward loop (Ras feed-forward).

The gap: every one of these is a fold-change or a single-edge knockdown. Nobody has asked, across the ladder, *which brakes are engaged in normal mucosa and released at which step, after confounders, and confirmed at the protein layer.* That is precisely the §6.4 join + §8 trajectory machinery — and CRC offers a **paired, multi-state, real-lesion ladder** that breast (where the DCIS arc had no within-patient stage pairs) structurally could not.

1.2 The single-cell atlases give the compartment map — but they are miRNA-*blind*

Two landmark atlases now chart polyp→CRC at cell-state resolution: **Chen et al., Cell 2021** (62 tumours / 128 datasets; conventional adenomas = WNT-driven stem-cell expansion, serrated polyps = gastric metaplasia) (Cell 2021) and **Becker, Curtis et al., Nat Genet 2022** (scATAC + scRNA over 48 polyps / 27 normal / 6 CRC; a stem-like continuum, FOX-regulated *pre-CAFs* maturing into RUNX1-CAFs, expanding Tregs, and T-cell exhaustion at the cancer step) (Nat Genet 2022).

These are **scRNA / scATAC / protein — there is no miRNA layer.** This is the single most important structural fact: the field has built the compartment/cell-state scaffold of CRC progression *with a miRNA-shaped hole in it.* The machine lives in exactly that hole, and the atlases supply the cell-state priors the composition covariates need (the pre-CAF→CAF and myeloid/Treg programs are the CRC analog of the CAF stroma-reframe the DCIS arc hit at MH-52/54).

1.3 Field cancerization: NAT is not neutral — and already carries prognostic miRNA signal

The "normal adjacent" tissue in CRC is a **field-cancerized** state, and miRNA dysregulation extends into it (field cancerization review). Concretely, a **four-miRNA coordinate deregulation in tumour-adjacent mucosa (miR-18a/miR-21/miR-182/miR-183) predicts relapse after curative resection** (NAT relapse predictor). This validates the decision to model NAT as an *intermediate trajectory state* rather than a clean baseline — and it means the acquired/field-effect classes have a ready-made outcome anchor in CRC.

1.4 Tissue miRNA–mRNA networks & TCGA: anticorrelation done naively

TCGA-COAD/READ has been mined repeatedly for miRNA–mRNA anticorrelation networks (hundreds of DE miRNAs, TargetScan/miRanda-filtered Pearson anticorrelation) (integrative knowledge base; regulatory network). These are **unweighted, unconfounded, single-edge** correlations — no evidence weighting, no promiscuity control, no purity/proliferation/CN adjustment, no permutation null. This is the exact methodology the framework was built to supersede; running the spine here is a direct, defensible upgrade with a known baseline to beat.

1.5 The protein layer: CPTAC-CRC exists, with the right shape, and is unused for pressure

Vasaikar et al., Cell 2019 is a **prospective 110-patient colon cohort with tumour, matched NAT, and blood**, profiled by WXS, CNV array, RNA-seq, **miRNA-seq, and label-free proteome (+phosphoproteome)** — the same data shape the §7 protein-realization layer already consumes for breast, with the bonus of **matched blood**. Proteogenomic discordance / post-transcriptional buffering in CRC is itself a documented phenomenon (mRNA–protein decoupling, 3′-UTR/APA effects) (post-transcriptional CRC review; APA dysregulation), which is exactly the "beyond-measured-mRNA residual" the §7 layer isolates. **Nobody has run evidence-weighted miRNA pressure → protein realization on CPTAC-CRC.** This is the cleanest *runnable-today* paper section and the non-circular positive control.

1.6 Circulating / EV biology: mechanistically rich, clinically incoherent

This is the loudest CRC literature and the messiest:

- **Selective EV sorting is a real, mechanistic phenomenon**, not random leakage: sumoylated **hnRNPA2B1** sorts specific miRNAs (GGAG/EXO-motif) into exosomes, **miR-1246** via APE1/hnRNPA2B1, and there is a **KRAS-dependent sorting program** (hnRNPA2B1 sorting, Nat Commun; KRAS-dependent secreted miRNA). So "which arm is exported" is *governed*, not noise — and the DCIS arc already built the transcriptional-loss vs vesicular-export distinction that this mechanism underwrites.
- **CAF/stromal exosomes** carry a defined CRC signature (miR-21 / miR-329 / miR-181a / miR-199b / miR-382 / miR-215; miR-92a-3p driving EMT/chemoresistance; miR-93-5p radioresistance) and concentrate in **CMS4** (CAF exosomal miRNA; miR-93-5p CAF). **Tumour exosomes build the liver pre-metastatic niche** (miR-25-3p vascular permeability, miR-934 M2 polarization, miR-188-3p hepatic stellate activation) (niche review).
- **Diagnostic panels are abundant but mutually contradictory** — the field's own reviews attribute the non-replication to **sample-type dependence** (free vs EV vs EPCAM⁺) (exosomal CRC diagnosis; combinatorial profiles), and the **preanalytical literature explains why translation has failed**: hemolysis distorts key arms (miR-451a), there is no consensus endogenous normalizer, and spike-ins don't rescue it (preanalytical challenges; hemolysis). Fecal miR-92a/miR-223 complement FIT for screening (fecal complement).

The gap is glaring: the circulating field has *mechanism* (selective sorting) and *clinical noise* (discordant panels) but **no model that links a circulating arm back to its tissue compartment's pressure/coupling state** and adjudicates *why* it is in blood (transcriptionally elevated in tumour vs selectively exported vs stromal-CAF-derived). That linkage is the thing the machine uniquely produces.

1.7 Subtype, immune, and outcome structure (the CRC-specific strata)

CRC's taxonomy is **CMS1-4** (MSI-immune / canonical-WNT-MYC / metabolic / mesenchymal-TGFβ), replacing PAM50 (CMS, Nat Med), with mature classifiers including a **miRNA-based one (CMS-miRaCl)** (CMS-miRaCl; CMScaller). miRNAs gate immune evasion (miR-148a-3p and miR-15b-5p → PD-L1, MSI-linked) (miRNA/PD-L1/MSI). On outcome, **tissue miR-21 is a robust prognostic marker** (OS HR ≈ 1.6–1.8 in meta-analysis) (miR-21 meta-analysis), while for **MRD/recurrence ctDNA dominates** (CIRCULATE-Japan / GALAXY) and circulating miRNA remains a promising-but-underpowered, mechanism-free add-on (ctDNA MRD; miRNA clearance panel).

1.8 One CRC-specific twist the gate should exploit: AGO2 is *lost*

Unlike the generic up-regulation of DROSHA/DICER in CRC, **AGO2 protein is reduced in CRC and predicts worse DFS/OS, driving EMT and metastasis** (AGO2/miR-185-3p/NRP1; Drosha/Dicer up). The §4.2 **availability gate** assumes RISC capacity (AGO/TNRC6) bounds realized repression — in CRC that assumption has a real, directional, prognostic substrate, so the gate is *more* mechanistically motivated here than in breast, and AGO2 loss becomes a finding to fold in rather than just a sensitivity layer.

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2. The white space, stated precisely

Putting the landscape together, every CRC sub-literature is missing the same thing, and it is the thing the machine *is*:

CRC literature has…	…but lacks (what the machine adds)
Exhaustive DE across the adenoma/serrated ladder	Potential×realized separation; confounder-adjusted **brake-release classification** along the ladder
Single-cell compartment/cell-state atlases (Chen; Becker-Curtis)	**Any miRNA layer** at compartment resolution — the atlases are scRNA/scATAC/protein
Naive TCGA miRNA–mRNA anticorrelation networks	Evidence weighting, promiscuity control, purity/proliferation/CN adjustment, permutation null, **protein realization**
CPTAC-CRC multi-omics incl. proteome	Evidence-weighted **pressure→protein** realization; the discordance is documented but unmodeled
Mechanistic EV sorting (hnRNPA2B1, KRAS) + discordant clinical panels	A model **linking a circulating arm to its tissue pressure state** and attributing it (transcriptional vs exported vs CAF)
ctDNA-led MRD; tissue miR-21 prognosis	A **mechanism-anchored** circulating signal tied to the tumour's realized repression field

And critically: the **two limits that closed the DCIS/EV arc** — no within-patient stage pairs, no compartment-resolved small-RNA — are **removed by the cohort described** (within-patient blood→polyp→NAT→tumour, with followup).

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3. How the machine maps onto CRC (the substrate port)

The framework is substrate-agnostic by design; only the cancer-flavoured priors change:

- **Strata:** CMS1-4 + MSI/MSS + sidedness + BRAF/KRAS, replacing PAM50. Classifiers (CMScaller, ColoType, CMS-miRaCl) are off-the-shelf.
- **Healthy anchor (§8.1):** GTEx **colon (sigmoid + transverse)** replaces GTEx breast.
- **Edge universe (§3):** miRTarBase / TargetScan / ENCORI are tissue-agnostic and transfer unchanged; CRC supplies *strong positive controls* (miR-21→PDCD4/PTEN, miR-143/145→KRAS, miR-17-92→targets, miR-34a→targets) to validate the spine recovers known biology unprompted, exactly as the breast oncomiR roster did.

- **Confounders (§6.2):** same stack; the **CMS4/desmoplasia stromal axis is the same stroma-reframe** the DCIS arc resolved (CAF activation), and **MSI/immune composition** is the CRC analog of the immune covariate. The Becker-Curtis/Chen cell-state programs feed composition deconvolution.
- **AGO gate (§4.2):** keep it, and additionally treat **AGO2 loss** as a directional CRC finding.
- **Program-polarity prior + gene-role layer (§5):** pan-cancer, transfer directly.

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4. Leverage map (what to run, in order)

Dataset	Role in CRC port	Status
TCGA-COAD/READ	Primary spine — miRNA+mRNA+CNV+methylation; CMS/MSI strata; beats the naive anticorrelation networks head-to-head	Runnable now
GTEx colon (sigmoid+transverse)	True-healthy anchor for the acquired axis	Runnable now
CPTAC-CRC (Vasaikar, 110; tumour+NAT+blood)	Protein-realization (§7) + NAT field state; the rigor anchor; matched blood is a bonus	Runnable now
Public EV/circulating + fecal GEO CRC	Replicate export-selectivity & the transcriptional-loss vs vesicular-export split before the new cohort	Runnable now
Chen 2021 / Becker-Curtis 2022 atlases	Cell-state composition priors + the miRNA-blind scaffold this layer complements	Reference now
New blood/polyp/NAT/tumour + followup cohort	The firsts in §5 — within-patient ladder, direct export mechanism, outcome-anchored pressure	When it lands

A full **pre-arrival validation** of the entire logic is runnable *today* on the first four rows — which de-risks the new cohort before a single sample exists.

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5. The headline the new cohort uniquely unlocks

> **A within-patient, compartment-resolved potential×realized "brake-release ledger" across the blood→polyp→NAT→tumour ladder, outcome-anchored by followup, in which every circulating arm is mechanistically attributed (transcriptional elevation vs selective EV export vs CAF-stromal origin) back to its tissue pressure/coupling state.**

This single design sits at the convergence of three gaps no existing CRC study closes simultaneously:

1. the premalignant-ladder DE literature has no **realization** model (the §6.4 brake-release classes on a *real, paired, dual-pathway* ladder);
2. the EV/circulating literature has no **tissue-of-origin mechanism** (the transcriptional-vs-export split, now testable *directly within patient* rather than triangulated across isogenic lines as in DCIS); and
3. the framework's own binding limits (no within-patient stage pairs, no compartment-resolved small-RNA) are **dissolved by the cohort design**.

It is also serrated-vs-conventional-pathway aware (a structure breast lacks entirely), giving a built-in second axis of biology the machine can resolve that no prior pressure analysis could.

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6. Four things to confirm before committing a roadmap

These change whether §5's components are "firsts" or "stronger replications":

1. **Blood modality** — EV / total small-RNA-seq vs a qPCR panel? (Determines whether the direct within-patient export-attribution test is possible at all.)
2. **Polyp↔tumour pairing** — same patients, or separate arms? (Determines within-patient ladder vs cross-sectional.)
3. **Pathway coverage** — are *both* conventional adenomas and serrated/SSL polyps sampled? (Unlocks the dual-pathway axis.)
4. **Followup endpoint** — recurrence/DFS vs OS, and is MRD/ctDNA co-collected? (Sets the outcome anchor and the ctDNA-complementarity framing.)